

Homework 3

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Exercise 1

Let the other player name an amount x . Note that the player can always get $10 - x$ by naming $10 - x$ as their amount since the total does not exceed 10. Suppose $x \leq 5$. Even if they name any y such that $x + y > 10$, since this implies that $y > x$, they will still receive $10 - x$. The best choice is then to name any $y \geq 10 - x$, receiving $10 - x$. Now suppose $x > 5$. If they name any amount between $10 - x$ and x , not including x , they will receive that amount. If they name x , they will receive 5. If they name any amount between x and 10, they will receive $10 - x$. The best choice is then to name $x - 1$, receiving $x - 1$ (except when $x = 6$, where they could also name 6).

x	Best y
0	$\{10\}$
1	$\{9, 10\}$
2	$\{8, 9, 10\}$
3	$\{7, 8, 9, 10\}$
4	$\{6, 7, 8, 9, 10\}$
5	$\{5, 6, 7, 8, 9, 10\}$
6	$\{5, 6\}$
7	$\{6\}$
8	$\{7\}$
9	$\{8\}$
10	$\{9\}$

	0	1	2	3	4	5	6	7	8	9	10
0											.
1										.	.
2									.	.	.
3								.	.	.	.
4							.	.	.	.	.
5						⊙	⊙	.	.	.	.
6					○	⊙	⊙	○			
7				○	○	○	.		○		
8			○	○	○	○		.		○	
9		○	○	○	○	○			.		○
10	○	○	○	○	○	○				.	

There are four Nash equilibria: $(5, 5)$, $(5, 6)$, $(6, 5)$, and $(6, 6)$.

Exercise 2

	L	C	R
T	0, 0	1, 0	1, 1
M	1, 1	1, 1	3, 0
B	1, 1	2, 1	2, 2

For Player 1, M weakly dominates T and B strictly dominates T . For Player 2, no action is weakly or strictly dominated. The only Nash equilibrium is (M, L) ; the equilibrium is not strict, since they would have the same payoff by switching to (M, C) or (B, L) .

Exercise 3

Suppose citizens are voting for k candidates using the approval voting system. Let α_i be the action where citizen i votes for their least preferred candidate (without loss of generality, let it be C_i). Consider the action α'_i , where the only difference is that citizen i does not vote for C_i . Either C_i is a winner with both α_i and α'_i , or A is a winner with α_i but not α'_i ; α'_i then weakly dominates α_i since C_i is citizen i 's least preferred candidate. Any action that includes a vote for a citizen's least preferred candidate is then weakly dominated; by symmetry, the same applies for any action that does not include a vote for a citizen's most favorite candidate.

Now let citizen i rank the k candidates as $1 > 2 > \dots > k$. Let α be the action where they vote only for candidate 1 and $k-1$. Now, let β be any action where citizen i does not vote for candidate 1, and consider the scenario where candidates 1 and 2 are tied as winners; switching to α would then have a better payoff since candidate 1 would be the winner, which means that α is not weakly dominated by β . Similarly, let γ be any action where citizen i does not vote for candidate $k-1$, and consider the scenario where candidates $k-1$ and k are tied as winners; switching to α would then have a better payoff since candidate k would not be a winner, which means that α is not weakly dominated by γ . Now, let δ be an action where citizen i votes for candidates 1 and $k-1$ in addition to some other candidate j ; and consider the scenario where candidates 1 and j are tied as winners; switching to α would then have a better payoff since candidate 1 would be the winner, which means that α is not weakly dominated by δ . Consequently, α is not weakly dominated by any action.